Equivariant Systems Theory and Observer Design for Autonomous Systems

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Outline

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Observer architecture

Recall the structure of the lifted system on the Lie group $\mathbf{G}$.

$$\dot{X} = F_{x_0}(X, v) = \mathrm{d}L_X \Lambda(\phi_X(x_0), v)$$
$$y_i = \rho^i(X, \mathring{y}_i) = \rho^i(X, h(x_0))$$

We propose to use the observer architecture

$$\dot{\hat{X}} = \mathrm{d}L_{\hat{X}} \Lambda(\phi_{\hat{X}}(x_0), v) + \mathrm{d}R_{\hat{X}} \Delta(\hat{X}, y_1, \ldots, y_p)$$
$$\hat{x}(t) = \phi_{x_0}(\hat{X}(t))$$

with initial condition $\hat{X}(0) \in \mathbf{G}$, typically $\hat{X}(0) = I$, and x_0 our best guess for $x(0)$. The correction term Δ remains to be designed.

Equivariant correction terms

We introduce the notations $\mathcal{Y} = \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_p$, $y = (y_1, \ldots, y_p)$ and $\mathring{y} = (\mathring{y}_1, \ldots \mathring{y}_p)$. We can also define the product action

$$\rho\colon \mathbf{G} \times \mathcal{Y} \to \mathcal{Y},\ (S, (y_1, \ldots, y_p)) \mapsto (\rho^1(S, y_1), \ldots, \rho^p(S, y_p))$$

An *equivariant correction term* is a correction term $\Delta \in \mathfrak{g}$ with

$$\Delta_{\mathring{y}}(R_S \hat{X}, \rho(S, y)) = \Delta_{\mathring{y}}(\hat{X}, y)$$

for all $S \in \mathbf{G}$ and $y \in \mathcal{Y}$. Note how we have made the parametric dependence of Δ on the reference measurements $\mathring{y}$ explicit in the notation.

Having Δ depend explicitly on time or velocity yields no additional advantage.

Observer problem

Kinematic system:

$$\begin{aligned} \dot{x} &= f(x, v) \\ y_i &= h_i(x) \end{aligned}$$

Lifted system:

$$\begin{aligned} \dot{X} &= \mathrm{d}L_X \Lambda(\phi_X(x_0), v) \\ y_i &= \rho^i(X, \mathring{y}_i) \end{aligned}$$

Observer:

$$\begin{aligned} \dot{\hat{X}} &= \mathrm{d}L_{\hat{X}} \Lambda(\phi_{\hat{X}}(x_0), v) + \mathrm{d}R_{\hat{X}} \Delta_{\mathring{y}}(\hat{X}, y) \\ \hat{x}(t) &= \phi_{x_0}(\hat{X}(t)) \end{aligned}$$

The goal: Find a correction term Δ such that some well chosen error variable has "nice" dynamics.

Outline

Equivariance and autonomy

A type I system is a kinematic system with a lifted system of the form

$$\dot{X} = \mathrm{d}L_X \Lambda(v).$$

Theorem: Consider a type I system and the observer as above. Then the dynamics of the canonical type I error $E_{\mathrm{I}}(\hat{X}, X) = \hat{X}X^{-1}$ is autonomous if and only if the correction term $\Delta_{\mathring{y}}(\hat{X}, y)$ is equivariant.

We can explicitly compute the error dynamics in this case.

$$\dot{E}_{\mathrm{I}} = -\mathrm{d}R_{E_{\mathrm{I}}} \Delta_{\mathring{y}}(E_{\mathrm{I}}, \mathring{y}).$$

(For type II systems, the appropriate error variable is the canonical type II error $E_{\mathrm{II}}(\hat{X}, X) = X^{-1}\hat{X}$).

The specific form of these error dynamics invites a **gradient based correction term design**.

Invariant cost functions

We start by defining *invariant cost functions* on the output spaces $\mathcal{Y}_i$.

$$\ell^i_{\mathring{y}_i} : \mathbf{G} \times \mathcal{Y}_i \to \mathbb{R}^+, \ (\hat{X}, y_i) \mapsto \ell^i_{\mathring{y}_i}(\hat{X}, y_i)$$

where

$$\ell^i_{\mathring{y}_i}(R_S \hat{X}, \rho^i(S, y_i)) = \ell^i_{\mathring{y}_i}(\hat{X}, y_i)$$

for all $S \in \mathbf{G}$. Then $\ell^i_{\mathring{y}_i}(\hat{X}, y_i) = \ell^i_{\mathring{y}_i}(E_{\text{l}}, \mathring{y}_i) = \ell^i_{\mathring{y}_i}(I, e^{\text{l}}_i)$ where $e^{\text{l}}_i = \rho^i(E_{\text{l}}^{-1}, \mathring{y}_i)$.

We require $(I, \mathring{y}_i)$ to be a global minimum of $\ell^i_{\mathring{y}_i}$. Moreover, we require that the cost is *non-degenerate* in the sense that $\mathrm{Hess}_2 \ell^i_{\mathring{y}_i}(I, \mathring{y}_i) > 0$.

Constructing invariant cost functions

Starting from a single variable cost function

$$f^i_{\mathring{y}_i} : \mathcal{Y}_i \to \mathbb{R}^+$$

with a global minimum at $y_i = \mathring{y}_i$, the construction

$$\ell^i_{\mathring{y}_i}(\hat{X}, y_i) := f^i_{\mathring{y}_i}(\rho^i(\hat{X}^{-1}, y_i)) = f^i_{\mathring{y}_i}(e^!_i)$$

yields an invariant cost function with global minimum at $(I, \mathring{y}_i)$ that is non-degenerate if $f^i_{\mathring{y}_i}$ is non-degenerate at $\mathring{y}_i$.

Aggregate cost

The aggregate cost across all outputs is

$$\ell_{\mathring{y}} \colon \mathbf{G} \times \mathcal{Y} \to \mathbb{R}^+, \; \ell_{\mathring{y}}(\hat{X}, y) = \sum_{i=1}^{p} \ell^i_{\mathring{y}_i}(\hat{X}, y_i) = \ell_{\mathring{y}}(E_{\mathrm{l}}, \mathring{y}) = \ell_{\mathring{y}}(I, e^{\mathrm{l}})$$

Lemma: $\ell_{\mathring{y}}$ is invariant and $\ell_{\mathring{y}}(\cdot, \mathring{y}) \colon \mathbf{G} \to \mathbb{R}^+$ has a global minimum at $E_{\mathrm{l}} = I$. It is non-degenerate if all the $\ell^i_{\mathring{y}_i}$ are non-degenerate and

$$\bigcap_{i=1}^{p} \mathrm{stab}_{\rho^i}(\mathring{y}_i) = \{I\}$$

For type I systems, the aggregate cost $\ell_{\mathring{y}}(\cdot, \mathring{y}) \colon \mathbf{G} \to \mathbb{R}^+$ can be directly used for observer design.

Gradient correction term

We define the *gradient correction term*

$$\Delta_{\mathring{y}}(\hat{X}, y) = k \mathrm{d}R_{\hat{X}^{-1}}(I) \operatorname{grad}_{\hat{X}} \ell_{\mathring{y}}(\hat{X}, y)$$

where the gradient is taken with respect to a right invariant Riemannian metric on **G**. The latter is contructed as $\langle \mathrm{d}R_S V, \mathrm{d}R_S W \rangle_S = \langle V, W \rangle$ where $V, W \in \mathfrak{g}$ and $\langle \cdot, \cdot \rangle$ is an inner product on $\mathfrak{g}$.

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \ell_{\mathring{y}}(E_{\mathrm{I}}, \mathring{y}) &= \sum_{i=1}^{p} \mathrm{d}_2 \ell^i_{\mathring{y}_i}(I, e^!_i) \mathrm{d}\rho^i_{e^!_i}(I)[\Delta] \\ &= -k \| \operatorname{grad}_{\hat{X}} \ell_{\mathring{y}}(E_{\mathrm{I}}, \mathring{y}) \|^2 \end{aligned}$$

Observer error convergence

Theorem: Consider a type I kinematic system and invariant cost functions $\ell^i_{\mathring{y}_i} : \mathbf{G} \times \mathcal{Y}_i \to \mathbb{R}^+$ with

$$\bigcap_{i=1}^{p} \mathrm{stab}_{\rho^i}(\mathring{y}_i) = \{I\}$$

Define the observer

$$\dot{\hat{X}} = \mathrm{d}L_{\hat{X}}\Lambda(v) - k\,\mathrm{grad}_{\hat{X}}\,\ell_{\mathring{y}}(\hat{X}, y)$$
$$\hat{x}(t) = \phi_{x_0}(\hat{X}(t))$$

Then there exists a basin of attraction $I \in \mathcal{B} \subset \mathbf{G}$ such that $E_\mathrm{l}(0) \in \mathcal{B}$ implies $E_\mathrm{l}(t) \to I$ and $\hat{X}(t) \to X(t)$.
If $\phi_{x_0}(X(0)) = x(0)$ then $\hat{x}(t) \to x(t)$.

Outline

Error dynamics revisited

For a kinematic system and observer as above, the *state error*

$$e = e(\hat{X}, x) := \phi(\hat{X}^{-1}, x)$$

has dynamics

$$\dot{e} = \mathrm{d}\phi_e \operatorname{Ad}_{\hat{X}} \left(\Lambda(\phi_{\hat{X}}(e), v) - \Lambda(\phi_{\hat{X}}(x_0), v) \right) - \mathrm{d}\phi_e \Delta_{\mathring{y}}(\hat{X}, y).$$

For an *equivariant* kinematic system this simplifies to

$$\dot{e} = \mathrm{d}\phi_e \left(\Lambda(e, v_t^0) - \Lambda(x_0, v_t^0) \right) - \mathrm{d}\phi_e \Delta_{\mathring{y}}(\hat{X}, y)$$

where $v_t^0 := \psi_{\hat{X}^{-1}}(v(t))$.

local coordinates and linearisation

Choose a local coordinate chart $\epsilon\colon \mathcal{U} \longrightarrow \mathbb{R}^m,\ e \mapsto \epsilon(e)$ with $x_0 \in \mathcal{U} \subset \mathcal{M}$ and $\epsilon(x_0) = 0$.

Choose a local coordinate chart $\delta\colon \mathcal{W} \longrightarrow \mathbb{R}^{n_1+\cdots+n_p}$ of the output (product) manifold $\mathcal{N}$ such that both $y \in \mathcal{W}$ and $\hat{y} \in \mathcal{W}$. Define $\tilde{\delta} := \delta(y) - \delta(\hat{y})$.

The linearised error dynamics around $e = x_0$ ($\epsilon = 0$) in these coordinates then have the form

$$\begin{aligned}\dot{\epsilon} &= A_t\epsilon + D\Delta\\ \tilde{\delta} &= C_t\epsilon\end{aligned}$$

where we might have to *switch* the coordinates δ whenever y or $\hat{y}$ moves too much.

local coordinates and linearisation

$$\begin{aligned} A_t &:= \mathrm{D}_{e|x_0}\varepsilon(e) \cdot \mathrm{D}_{E|\mathrm{id}}\phi_{x_0}(E) \cdot \mathrm{D}_{e|x_0}\Lambda(e, v_t^0) \cdot \mathrm{D}_{\varepsilon|0}\varepsilon^{-1}(\varepsilon), \\ C_t &:= \mathrm{D}_{y|y_t}\delta(y) \cdot \mathrm{D}_{x|\hat{x}}h(x) \cdot \mathrm{D}_{e|x_0}\phi_{\hat{X}}(e) \cdot \mathrm{D}_{\varepsilon|0}\varepsilon^{-1}(\varepsilon), \\ D &:= \mathrm{D}_{e|x_0}\varepsilon(e) \cdot \mathrm{D}_{E|\mathrm{id}}\phi_{x_0}(E) \end{aligned}$$

If the output is *equivariant* and allows for a single output manifold chart then $C_t \equiv C$ is *constant*! In this case

$$C = \mathrm{D}_{d|\mathrm{y}}\delta(d) \cdot \mathrm{D}_{e|x_0}h(e) \cdot \mathrm{D}_{\varepsilon|0}\varepsilon^{-1}(\varepsilon)$$

where $d = e^{\mathrm{l}} = \rho_{\hat{X}^{-1}}(y)$.

The Equivariant Filter (EqF)

The EqF is based on a Riccati observer design

$$\begin{aligned}
\dot{\hat{X}} &= \mathrm{d}L_{\hat{X}}\Lambda(\phi_{x_0}(\hat{X}), v) \\
&\qquad - \mathrm{d}R_{\hat{X}}\mathrm{d}\phi^{\dagger}_{x_0}\mathrm{D}_{\varepsilon|0}\varepsilon^{-1}(\varepsilon)\Sigma {C^t}^{\top} N_t^{-1}\left[\delta(y) - \delta(h(\phi_{x_0}(\hat{X})))\right] \\
\dot{\Sigma} &= A_t\Sigma + \Sigma A_t^{\top} + M_t - \Sigma {C_t}^{\top} N_t^{-1} C_t \Sigma, \\
\hat{x}(t) &= \phi_{x_0}(\hat{X}(t))
\end{aligned}$$

with initial conditions $\hat{X}(0) \in \mathbf{G}$ (usually $\hat{X}(0) = I$) and $\Sigma(0) = \Sigma_0$ positive definite, and gain matrices M_t and N_t.

EqF error convergence

Theorem: Consider an equivariant kinematic system with equivariant output and equivariant lift. Let M_t and N_t be positive definite symmetric gain matrices with

$$\lambda I_m < M_t$$

for a constant $\lambda > 0$.
If solutions $(\hat{X}(t), \Sigma(t))$ to the equivariant filter equation exist for $t \to \infty$ and

$$\sigma_1 I_m < \Sigma(t) < \sigma_2 I_m$$

for constants $\sigma_2 > \sigma_1 > 0$ then the state error $e = \phi(\hat{X}^{-1}, x)$ is uniformly locally exponentially stable in a neighbourhood of $e = x_0$.

Outline

1. Observers
2. Gradient observer design
3. EqF design
4. Conclusion

Examples that we covered

Attitude observer: $\dot{R} = R\Omega_\times$, Type I system on $\mathrm{SO}(3)$, gradient error dynamics (*beautiful!!!*)

Attitude observer with bias: extensions to that theory with Lyapunov analysis (*cool!!!*)

Homography observer: $\dot{H} = H(\Omega_\times + \Gamma)$, *not* Type I (because Γ depends on state information), but we model as Type I with bias (*works!!!*)